



AQ4.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

Find the rank of the following matrix: $\begin{bmatrix} 0 & 1 & 3 & 1 \\ 1 & 0 & 0 & 1 \\ 3 & 2 & 0 & 3 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Consider the following set of vectors SS in R^3 to answer the questions 2, 3 and 4.

$S = \{(1, 2, 0), (0, 3, 1), (3, 3, -1), (3, 0, -2)\}$

1 point

Let V be the vector space spanned by the vectors $(1, 2, 0)$ and $(0, 3, 1)$. Which one of the following is correct?

☐

$V = \{(a, 2a + 3b, b) \mid a, b \in R\}$

☐

$V = \{(0, 2a - 3b, b) \mid a, b \in R\}$

☐

$V = \{(-a, 2a + 3b, 0) \mid a, b \in R\}$

☐

$V = \{(0, 2a + 3b, 0) \mid a, b \in R\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \{(a, 2a + 3b, b) \mid a, b \in R\}$

If the vectors in SS are written as the columns of a matrix A , then what will be the rank of A ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

What will the dimension of the vector space spanned by SS be?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point



1 point

Choose the set of correct options from the following.

- ☐ Row rank and column rank of a matrix is always the same.
- ☐ The rank of a zero matrix is always 0.
- ☐ The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- ☐

Rank of the $n \times n$ identity matrix is 1 for any $n \in \mathbb{N}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Row rank and column rank of a matrix is always the same.

The rank of a zero matrix is always 0.

The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

Level 2

1 point

Consider the following upper triangular matrix to choose the correct options.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

where $a, b, c, d, e, f \in \mathbb{R}$.

☐ If $f = 0$, then the rank of the matrix must be less than or equal to 2.

☐ If $f = 0$, then the rank of the matrix must be exactly 2.

☐ If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.

☐ If a, d, f are non-zero then the rank of the matrix must be 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $f = 0$, then the rank of the matrix must be less than or equal to 2.

If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.

If a, d, f are non-zero then the rank of the matrix must be 3.

If rank of the matrix $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$ is 2 then find the value of a .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

Find the rank of the matrix A , where $A = [a_{ij}]$ is of order 3×3 and $a_{i,j} = \min\{i, j\}$, $i, j = 1, 2, 3$.

[Hint: Write down the matrix and apply row operations to derive its reduced row echelon form.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

What is the rank of the matrix $A = \begin{bmatrix} x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$, where x, y, z are distinct non-zero integers?

non-zero integers?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Rank of a 3×3 matrix A is 3 and another 3×3 matrix B is 2. What is the minimum possible value for the rank of $ABAB$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

[Check Answers](#)

Your score is: 0/10